Deep Kabariya

AI/ML Engineer | NLP | AI Systems | Research & Analysis

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LinkedIn · GitHub · Portfolio · LeetCode

Profile

Computer Engineering student with hands-on experience building and evaluating AI systems involving natural language processing, machine learning, semantic retrieval, recommendation systems, and AI-assisted evaluation. Experienced in working with structured and unstructured data, designing end-to-end analytical pipelines, evaluating model outputs, and documenting technical workflows. Particularly interested in applying AI, data analysis, and emerging-technology research to strategic foresight, horizon scanning, and evidence-based decision support.

Education

Bachelor of Engineering – Computer Engineering *Expected 2027*

Shree Swami Atmanand Saraswati Institute of Technology (SSASIT), Surat CGPA: 8.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Theory of Computation

DSA Practice: Regular problem solving in Java and Python with solutions documented in a public repository.

Research & Technical Skills

AI & Machine Learning: Generative AI, Large Language Models, Machine Learning, Natural Language Processing, Deep Learning, Data Science

NLP & Information Analysis: Text Classification, Text Preprocessing, Tokenization, Embeddings, Semantic Search, Cosine Similarity, Information Retrieval, Natural Language Understanding

AI-Assisted Systems: LLM Evaluation, AI Output Quality Assurance, Prompt-Based Workflows, Multi-Provider Orchestration, Human Review Workflows, RAG-Style Retrieval, AI Reliability

Research Methods & Analysis: Problem Formulation, Experimental Methodology, Comparative Evaluation, Data Cleaning, Data Validation, Qualitative Result Analysis, Technical Documentation

Frameworks & Libraries: TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers, LangChain, LlamaIndex, NumPy, Pandas, Streamlit

AI Platforms & APIs: Google Gemini, Groq, GPT Models, Azure OpenAI Service, Gemini Embeddings, OpenAI-Compatible APIs

Programming & Infrastructure: Python, JavaScript, TypeScript, Java, SQL, Node.js, Express, MongoDB, PostgreSQL, Redis, Git, Docker, Linux

Experience

1Stop.ai *Mar 2025 – Apr 2025*

Artificial Intelligence Intern Remote

- Developed Python automation for structured data ingestion, cleaning, validation, and processing within an AI engineering workflow.
- Quality-controlled and structured **100,000+ records** to meet data requirements for machine learning workflows.
- Documented API architectures and system workflows, contributing to a **25% reduction** in developer onboarding time.
- Worked within an AI engineering environment and gained practical exposure to data quality, technical documentation, and production-oriented AI workflows.

Stack: **Python** · **Pandas** · **NumPy** · **Data Pipelines** · **REST APIs**

Certificate · Verify

CoBalance – Fintech System Design *Jan 2026 – Present*

Independent Research & Engineering

- Investigated reliability and security patterns for financial data systems, including token rotation, ownership validation, row-level security, and atomic transaction design.
- Applied systematic performance profiling to identify database bottlenecks and reduced API response latency by **40%** through targeted indexing.
- Documented technical architecture and evaluated implementation trade-offs involving security, data integrity, performance, and concurrent operations.

Research Projects

CineStream – Hybrid Recommendation & Semantic Retrieval Live · GitHub

- Investigated whether combining content-based filtering, collaborative signals, and semantic vector search could improve recommendation relevance compared with single-signal approaches.

- Generated catalogue embeddings and used cosine similarity for content-based retrieval, while interaction history contributed collaborative signals for personalization.
- Implemented semantic search using MongoDB Atlas Vector Search, allowing natural-language queries to retrieve conceptually relevant information.
- Built a RAG-style conversational workflow where retrieved catalogue information is supplied to the LLM before response generation, grounding outputs in available data.

Stack: **Gemini Embeddings** · **MongoDB Atlas Vector Search** · **Redis** · **Node.js** · **React**

SkillSync – AI-Based Competency Evaluation Live · GitHub

- Designed an AI-assisted competency evaluation workflow in which generated challenges are assessed against structured rubrics before verified badges are issued.
- Built natural-language processing workflows that decompose project descriptions into technical skills and required team roles.
- Implemented multi-provider AI orchestration using Gemini as the primary provider with Groq fallback for handling provider errors and timeouts.
- Explored AI reliability through structured evaluation rules and provider-level failure handling.

Stack: **Gemini** · **Groq** · **NLP** · **Next.js** · **TypeScript** · **MongoDB**

SMS Spam Classifier – NLP Classification Study GitHub

- Designed an end-to-end NLP classification pipeline using the SMS Spam Collection benchmark dataset containing **5,572 labeled messages**.
- Applied text normalization, tokenization, out-of-vocabulary handling, sequence padding, trainable embeddings, and LSTM-based classification.
- Evaluated model performance using validation accuracy, precision, and recall, achieving **98.6% validation accuracy, 99% precision, and 96% recall**.
- Deployed the trained model through Streamlit for real-time inference with confidence scores.

Stack: **Python** · **TensorFlow** · **Keras** · **NLP** · **Streamlit**

Real-Time Voice AI Pipeline – Speech-to-Speech Agent GitHub

- Designed a modular VAD → STT → LLM → TTS pipeline for low-latency human-AI speech interaction.
- Compared speech recognition and synthesis components based on speed, accuracy, multilingual capability, output quality, and deployment requirements.
- Exposed the pipeline through a WebSocket interface and evaluated local deployment options for privacy-sensitive use cases.

Stack: **Python** · **Silero VAD** · **Whisper** · **Parakeet TDT** · **Qwen3-TTS** · **WebSocket**

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31 August 2026

Hiring Team
United Nations Population Fund (UNFPA)
Strategic Planning Branch, Programme Division

Subject: Application for Foresight Intern

Dear Hiring Team,

I am writing to apply for the Foresight Intern position with UNFPA's Strategic Planning Branch. I am currently pursuing a Bachelor's degree in Computer Engineering and have focused much of my practical work on artificial intelligence, natural language processing, data analysis, and the evaluation of AI-based systems. The opportunity to apply these technical skills to strategic foresight and long-term decision-making is particularly interesting to me because it brings together two areas I want to understand more deeply: emerging technologies and the way organizations make decisions under uncertainty.

Through my academic and project work, I have built and evaluated AI systems that work with large amounts of information and natural language. My projects include an NLP classification system, a hybrid recommendation platform combining semantic search with recommendation methods, and an AI-based competency evaluation platform with rubric-based assessment and multi-provider model orchestration. These projects have required me to think about data quality, classification, retrieval, model behaviour, evaluation, and the reliability of AI-generated outputs rather than treating an AI model as a black box.

I am particularly interested in the AI-assisted component of this internship. The use of AI for horizon scanning, signal classification, clustering, and analytical support raises practical questions about evaluation, reliability, human review, and how emerging signals can be turned into useful information for decision-makers. My experience building fallback mechanisms between AI providers and designing automated evaluation workflows has given me an appreciation for these issues from a technical perspective. I would like to develop that perspective further by learning how such tools can support foresight work in an international development context.

I am also drawn to the opportunity to learn from UNFPA's work on demographic change, strategic planning, anticipatory governance, and adaptive programming. While my background is in engineering rather than a traditional policy discipline, I believe that perspective can be useful in a multidisciplinary environment where technology, data, uncertainty, and long-term development challenges increasingly intersect. I am comfortable learning unfamiliar domains, working independently on technical problems, documenting my work, and collaborating with others to turn complex information into practical outputs.

I would value the opportunity to contribute my technical and analytical background while developing a stronger understanding of strategic foresight and its application within the United Nations system. Thank you for considering my application. I would be grateful for the opportunity to discuss how my background and interests could contribute to the Foresight team's work.

Kind regards,

Deep Kabariya
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